

Training-Free Generation of Protein Sequences from Small Family Alignments via Stochastic Attention

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1 Generating novel protein sequences that respect the statistical constraints of a family typically requires training deep generative models
2 on thousands to millions of examples. Yet most protein families are small: the median Pfam seed alignment contains only 22 sequences,
3 a regime where learned models overfit or collapse. We propose *stochastic attention* (SA), a training-free sampler that treats the modern
4 Hopfield energy over stored sequences as a Boltzmann distribution and draws samples via Langevin dynamics. The score function is given in
5 closed form by the residual of a single softmax attention operation, eliminating the need for a trained score network, external pretraining
6 data, or graphics processing unit (GPU) resources. Across eight Pfam families spanning 37 to 420 sequences and 23 to 262 residues, SA
7 generates sequences with low amino acid composition divergence, substantial novelty, and structural plausibility confirmed by ESMFold
8 and AlphaFold2. Head-to-head comparison with profile HMMs, EvoDiff, and the MSA Transformer shows that SA is the only method to
9 simultaneously achieve low composition divergence, genuine novelty, and sequence identity within the natural within-family range (30–70%);
10 competing methods either drift outside the family or produce near-copies of stored patterns. The critical inverse temperature is predicted from
11 principal component analysis (PCA) dimensionality alone, enabling fully automatic operation from a seed alignment. Independent validation
12 by ESM2-650M and cross-referencing against deep mutational scanning experiments confirm that over 90% of SA-generated substitutions
13 correspond to experimentally tolerated mutations. Stochastic attention thus opens training-free sequence generation to the small families that
14 constitute most of the protein universe.

protein sequence generation | stochastic attention | Hopfield networks | Langevin dynamics | training-free generative models

1 With protein structure prediction largely solved by AlphaFold2 (1) and RoseTTAFold (2), the frontier
2 has shifted toward *designing* novel protein sequences with desired properties (3, 4). Generative
3 approaches now span a broad landscape, from classical profile hidden Markov models (pHMMs) (5, 6) and
4 Potts models (7–9) to deep generative models including variational autoencoders (10, 11), autoregressive
5 language models (12, 13), alignment-processing transformers (14), MSA-conditioned diffusion (15), and
6 structure-conditioned methods such as ProteinMPNN (16) and RFdiffusion (17). Yet all sequence-level
7 approaches share a common requirement: large training corpora. Language models are pretrained on
8 millions of UniRef50 sequences; even MSA-conditioned methods require either extensive pretraining or large
9 input alignments. This creates a gap for small families: the median Pfam seed alignment contains only 22
10 sequences, and nearly half of all families have fewer than 20 (SI Appendix, Section S21), well below the
11 minimum viable training set for any deep generative architecture. In this scarce-data regime, deep models
12 overfit and collapse in diversity, while pHMMs emit sequences that drift far from the family’s functional
13 envelope. What is needed is an approach that can extract the statistical structure of a protein family from
14 whatever data is available, even a few dozen sequences, without learning parameters that could overfit.

15 A separate line of work provides exactly this capability. Ramsauer et al. (18) showed that a single softmax
16 attention operation is equivalent to one step of the retrieval dynamics on the modern Hopfield energy,
17 building on the dense associative memory framework of Krotov and Hopfield (19). This connection endows
18 any set of stored examples with a well-defined energy landscape: the examples become memory patterns,
19 and the Boltzmann distribution over this landscape assigns high probability to states that resemble the
20 stored set. The energy itself has a transparent structure: a quadratic term $\frac{1}{2}\|\xi\|^2$ pulls the state toward the
21 origin, while a log-sum-exp term $-\frac{1}{\beta}\log\sum_k\exp(\beta\mathbf{m}_k^\top\xi)$ pulls it toward stored memories. The gradient of
22 the log-sum-exp is precisely the softmax-weighted combination of stored patterns, i.e., a single attention
23 operation. The score function of the Boltzmann distribution is therefore the *residual* of attention: the
24 difference between the attention output and the current state. This residual is exact and available in

closed form, so Langevin sampling can proceed without learning any parameters, training a score network, or approximating gradients. We exploit this insight to propose *stochastic attention* (SA), a training-free protein sequence generator that combines the exact Hopfield score function with Langevin dynamics. Our prior work (20) introduced SA and validated it on synthetic data, MNIST, and a single protein family.

In this study, we explored the extent to which SA can serve as a practical generative model for real protein families. We systematically evaluated SA across eight diverse Pfam families spanning a ten-fold range of family sizes and a seven-fold range of sequence lengths, assessing compositional fidelity, sequence novelty, and structural plausibility using both ESMFold and AlphaFold2. We characterized how generation quality scales with family size, uncovering an empirical law that predicts the critical temperature from alignment statistics alone and eliminates the need for a temperature sweep. We tested robustness in the extreme scarce-data regime down to $K=20$ sequences, compared SA head-to-head against four established baselines (profile HMMs, EvoDiff, MSA Transformer, and a Potts model), and examined whether the generated sequences preserve the pairwise residue covariation that reflects structural contacts and functional couplings. Finally, we sought independent validation of biological plausibility through ESM2-650M pseudo-perplexity scoring and cross-referencing against published deep mutational scanning datasets. Across all eight families, SA generated novel sequences that preserved family-level amino acid statistics and were predicted by both ESMFold and AlphaFold2 to adopt the correct fold, a combination no learned baseline matched. Together these results indicate that the statistical structure of a protein family can itself support sequence generation, without learned representations, when the sampler respects the geometry of the family's sequence space.

Results

We ran SA on all eight Pfam families (Table 1) and observed that the method consistently generated novel protein sequences while preserving family-level amino acid statistics, across a ten-fold range of family sizes and a seven-fold range of sequence lengths (Fig. 1; SI Appendix, Table S1). We first confirmed that the sampler converged to the correct Boltzmann distribution by comparing the outputs of the unadjusted Langevin algorithm (ULA) and its Metropolis-adjusted variant (MALA): acceptance rates exceeded 99.6% in every family (SI Appendix, Table S8), confirming negligible discretization bias and justifying the use of the computationally cheaper ULA in all subsequent experiments. Generated sequences were not near-duplicates of stored patterns: zero sequences across all eight families exceeded 90% identity to any stored sequence (SI Appendix, Table S10), and the generated ensemble was insensitive to initialization strategy (SI Appendix, Table S9). In the generation regime (inverse temperature $\beta \approx 2\beta^*$, just above the entropy inflection point where pattern-specific basins first emerge), SA achieved Kullback–Leibler (KL) divergences below 0.06 in every family, with several below 0.02. Novelty ranged from 0.40 (Defensin_beta) to 0.65 (WW), indicating

Significance Statement

Most protein families are small, with a median of only 22 sequences in their Pfam seed alignments, too few to train a deep generative model. We show that a single attention operation over a family alignment defines an energy landscape whose Boltzmann distribution can be sampled by Langevin dynamics without training, GPUs, or external data. The resulting method, stochastic attention, generates novel protein sequences that preserve family-level amino acid statistics and are predicted by independent structure predictors to fold correctly across eight diverse families. A simple linear relationship between representation dimensionality and the critical sampling temperature enables fully automatic operation from a seed alignment, opening training-free generation to the majority of protein families too small for deep learning.

J.D.V. designed research, performed research, analyzed data, and wrote the paper.

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substantial departure from any single stored memory, and mean pairwise cosine distance was ≈ 1.0 in every family, confirming that the sampler produced distinct, well-separated sequences rather than collapsing to a single mode. Nearest sequence identity to a stored pattern was 51–66%, comparable to the typical within-family identity range, indicating that generated sequences remained within the family’s functional envelope while departing meaningfully from stored patterns. By contrast, bootstrap samples had zero novelty by construction and Gaussian perturbation produced negligible novelty (< 0.02), confirming that small additive noise is insufficient to escape the basin of the nearest stored pattern.

Position-specific analysis of the Kunitz domain revealed why SA preserves family statistics (SI Appendix, Fig. S5). SA-generated sequences preserved all 11 positions with greater than 90% conservation in the seed alignment, including all six cysteines forming the three disulfide bonds essential to the Kunitz fold, while introducing 17–25 substitutions concentrated at variable positions. Per-position Shannon entropy correlated strongly between the SA-generated ensemble and the stored alignment ($r=0.968$), confirming that the method correctly identified which positions tolerate variation and which do not. This behavior followed directly from the Boltzmann sampling mechanism: at strongly conserved positions, all stored patterns agreed and the softmax weights concentrated probability on the consensus residue; at variable positions, stored patterns diverged and the sampler drew from the local distribution of tolerated amino acids.

Subsampling the WW domain at $K \in \{20, 50, 100, 200, 400\}$ with five independent replicates per condition characterized how generation quality changed with decreasing family size (SI Appendix, Fig. S2). Stochastic attention maintained low KL divergence across the entire range (0.019 ± 0.008 at $K=20$, improving to 0.010 ± 0.002 at $K=400$), while the PCA dimensionality grew from $d=17$ –18 components at $K=20$ to $d=182$ –183 at $K=400$ and the critical temperature tracked this growth ($\beta^*=2.7$ to 5.5). Despite the roughly ten-fold increase in effective dimensionality, SA generation quality remained stable, suggesting that the entropy inflection criterion successfully adapted the operating temperature to the geometry of the memory matrix at each scale. Novelty increased with K (from 0.47 to 0.74), reflecting the richer landscape of a larger memory set, while nearest sequence identity decreased correspondingly (from 0.71 to 0.56), indicating that the sampler explored further from stored patterns when more patterns were available to collectively shape the energy landscape. Replication across three additional families at $K=20$ (SH3, Kunitz, zf-C2H2) confirmed that this robustness is not family-specific (SI Appendix, Table S3).

The scaling study revealed that β^* tracked PCA dimensionality, prompting us to ask whether this relationship could predict β^* from alignment statistics alone, without a temperature sweep (Fig. 2). We pooled 33 data points (the 8 Pfam families and 25 WW scaling replicates) and found that a simple two-parameter model, $\beta^* \approx 1.52 + 0.28\sqrt{d}$ (bootstrap standard error [SE]: intercept ± 0.07 , slope ± 0.007), explained 97% of the variance ($R^2=0.97$). The intercept of 1.52 recovers the universal softmax inflection baseline, and the fitted slope of 0.28 falls well below the $\sqrt{d}$ coefficient of ≈ 1.6 predicted for random patterns. This reduction arises from two effects: concentration of measure on $\mathbb{S}^{d-1}$ guarantees that cross-similarities between stored patterns scale as $\mathcal{O}(1/\sqrt{d})$, setting the $\sqrt{d}$ dependence, while self-similarity anchoring (each stored pattern has unit inner product with itself, creating a gap of 0.55–0.82 over the next-nearest pattern) shifts the inflection to lower β than a Gaussian mean-field model predicts; together these effects reflect the structured similarity spectrum of real protein sequences arising from conserved positions, correlated substitutions, and phylogenetic signal (SI Appendix, Section S19). Adding further predictors (mean column entropy, effective number of sequences, spectral concentration) produced negligible improvement ($\Delta R^2 < 0.001$), indicating that PCA dimensionality captures most of the relevant variation. Refitting on the eight Pfam families alone yielded a nearly identical relationship ($R^2=0.95$), confirming that the result is not driven by overrepresentation of the WW family. This regression provides a practical shortcut: given a new family, one can estimate β^* from PCA alone and set the generation temperature without a sweep.

We compared SA against three established learned baselines, profile HMM emit, EvoDiff (MSA-conditioned diffusion), and the MSA Transformer (Gibbs masked language model sampling), across all eight families (Fig. 3). Reading the three panels jointly is essential: no single metric tells the full story, and a method

that scores well on novelty while failing on compositional fidelity or sequence identity has simply generated sequences that no longer belong to the family. Profile HMM emit and the MSA Transformer achieved high novelty scores, but this apparent advantage is illusory: their nearest sequence identity to stored patterns fell to 9–59% in several families, well below the typical within-family range of 30–70% (shaded band, Fig. 3C). Sequences below this range have drifted outside the family and are unlikely to retain its function; sequences above it are near-copies of stored patterns with insufficient novelty. The high novelty of HMM emit and the MSA Transformer reflects compositional drift, not meaningful exploration of family sequence space. EvoDiff balanced the metrics more effectively in shorter families but required pretraining on millions of UniRef50 MSAs, scaled poorly with sequence length (~14 hours for 150 Pkinase sequences on CPU, though GPU inference would reduce this substantially, versus seconds for SA on the same hardware; SI Appendix, Table S11), and suffered a sharp degradation in compositional fidelity on the longest family (Pkinase KL=0.17, more than an order of magnitude above its values in the other seven families).

Stochastic attention was the only method that simultaneously achieved all three criteria (Fig. 3). It held KL divergence below 0.05 in seven of eight families (a KL of 0.05 over 20 amino acids corresponds to an average absolute frequency error of less than 0.5 percentage points per residue type, well within finite-sample variation), produced novelty of 0.40–0.65 indicating genuine departure from stored patterns (zero generated sequences exceeded 90% identity to any stored pattern, and zero were within two substitutions of a stored sequence across all eight families), and kept sequence identity at 51–66%, firmly within the natural within-family range. This combination (compositionally faithful, genuinely novel, and within the family) distinguished SA from every baseline tested, using only the seed alignment and no pretraining, backpropagation, or GPU resources. A systematic comparison with a Potts model fitted via pseudo-likelihood maximization (plmDCA) across all eight families (SI Appendix, Table S7) provided a more informative contrast, because Potts models are the most natural physics-based competitor: they explicitly fit pairwise coupling parameters $J_{ij}(a, b)$ between all position pairs, giving them direct access to coevolutionary signal. The comparison revealed comparable compositional fidelity and broadly similar output quality in six of eight families, with SA achieving higher novelty in five of eight. However, the Potts model showed clear failure modes in the most data-starved families (Pkinase: novelty of 0.22 vs. 0.48 for SA, indicating the Gibbs sampler remained trapped near stored patterns; Defensin_beta: KL=0.112, the worst of any method), consistent with its extreme over-parameterization: the number of coupling parameters scales as $L(L-1)q^2/2$, yielding data-to-parameter ratios as low as 2.5×10^{-6} . That SA achieves comparable or better output without fitting any parameters suggests that the Hopfield energy implicitly captures the correlational structure that pairwise coevolutionary models fit explicitly, but through the softmax attention operation rather than through parameter estimation. Whereas the Potts model encodes pairwise couplings, the softmax nonlinearity in the Hopfield energy couples all positions simultaneously through the attention weights, effectively encoding all-order correlations without the combinatorial explosion of explicit higher-order terms.

Two control experiments confirmed that SA captured more than marginal per-position amino acid frequencies (SI Appendix, Tables S12 and S13). First, a consensus-with-noise baseline that added calibrated isotropic Gaussian perturbations in PCA space produced a qualitatively different output profile: systematically higher novelty (0.63–0.79 vs. 0.40–0.65 for SA) paired with lower sequence identity to stored patterns (0.44–0.62 vs. 0.52–0.66), reflecting the fact that isotropic noise scattered sequences away from all stored patterns rather than interpolating between them in a structured way. Second, SA run on column-permuted alignments, which preserved the exact per-position amino acid frequencies but destroyed all inter-position correlations, yielded consistently different output: higher novelty and lower sequence identity than SA on the real alignment in all eight families, with MI correlation differences favoring the real alignment in seven of eight families. The larger differences in novelty and sequence identity confirm that inter-position correlations present in real alignments constrain the energy landscape in ways that marginal frequencies alone cannot reproduce, and that SA exploits this correlational structure during generation.

The computational cost of SA differed from learned baselines by orders of magnitude (SI Appendix,

Table S11). SA generated 150 sequences in 0.2–4.6 seconds across all eight families on a single CPU core, with the variation driven primarily by the number of stored patterns K (which determines the cost of the softmax attention at each Langevin step). Profile HMMs were consistently the fastest method (0.02–0.09 seconds), as expected for sampling from a precomputed position-specific model. EvoDiff required 2–14 hours on CPU depending on sequence length, and the MSA Transformer required 1–8 hours for iterative masked language model sampling. Potts model fitting and Gibbs sampling required 1–52 minutes, with the Pkinase family ($L=262$) dominating due to the $\mathcal{O}(L^2q^2)$ parameter count. These comparisons confirm that SA occupies a qualitatively different computational regime from learned baselines: it is 3–5 orders of magnitude faster than EvoDiff and the MSA Transformer, requires no GPU, no pretraining data, and no parameter optimization. This accessibility is not incidental; it is a direct consequence of the closed-form score function, which eliminates the need for backpropagation, gradient accumulation, or iterative parameter updates.

We assessed structural plausibility by predicting folded structures for 50 sequences per method per family using ESMFold (Fig. 4), comparing SA generation to stored natural sequences via Wilcoxon rank-sum tests with Cohen’s d effect sizes. SA-generated sequences achieved mean predicted local distance difference test (pLDDT) scores statistically indistinguishable from stored sequences in six of eight families ($p > 0.05$), with significantly higher pLDDT in Kunitz (90.8 vs. 89.3, $p=0.001$, $d=0.67$), and at least 86% of SA-generated sequences exceeded the pLDDT=70 confidence threshold in seven of eight families, with SH3 the lone exception (54%). Template modeling (TM) scores showed a consistent pattern: in six of eight families, SA generation achieved significantly higher TM-scores than stored sequences ($p < 0.05$; $d > 1.0$ in RRM, WW, PDZ, and Pkinase), likely reflecting the Boltzmann ensemble’s tendency to weight the central tendency of the family distribution, producing sequences closer to a consensus fold than individual natural sequences, which carry lineage-specific structural deviations. As an independent cross-validation, we repeated the structure prediction for all eight families using AlphaFold2 (SI Appendix, Figs. S3–S4). The AF2 results corroborated the ESMFold findings while resolving the single negative result: for SH3, the only family where ESMFold had indicated a pLDDT deficit, AlphaFold2 showed no pLDDT difference ($p=0.99$) and significantly higher TM-scores for SA generation (0.736 vs. 0.721, $p < 0.001$). The concordance between two independent structure predictors across all eight families strengthens the computational evidence that SA-generated sequences are predicted to adopt the correct fold (Fig. 5).

Pairwise mutual information (MI) analysis tested whether SA preserved residue covariation, which reflects structural contacts and functional couplings that single-site statistics cannot capture (SI Appendix, Table S4; Figs. S6–S7). Pearson correlations between stored and SA-generated MI vectors ranged from $r=0.53$ to 0.92 across seven of eight families, with Pkinase ($r=0.05$) the lone exception, owing to its extreme data-to-position ratio ($K=37$, $L=262$). SA consistently exceeded profile HMMs ($r=0.07$ –0.83), and the general ordering $\text{HMM} < \text{SA} < \text{Potts} \leq \text{EvoDiff}$ held in three of eight families (with SA above HMM in seven of eight), consistent with the information hierarchy of each approach: position-independent frequencies, implicit all-order Hopfield couplings, explicitly fitted pairwise couplings, and deep coevolutionary priors from large-scale pretraining, respectively. This ordering confirmed that the Hopfield energy captured correlational structure beyond what position-independent models encode, without requiring explicit pairwise parameter estimation.

As a complementary, sequence-level assessment of biological plausibility, we scored all SA-generated and stored sequences using ESM2-650M (21), a protein language model pretrained on millions of UniRef50 sequences that is entirely independent of SA (SI Appendix, Table S5). SA-generated sequences were statistically indistinguishable from natural family members in four of eight families (RRM, zf-C2H2, Pkinase, Defensin_beta; Wilcoxon $p > 0.29$), with SA achieving significantly *better* pseudo-perplexity in the Kunitz domain (4.47 vs. 4.98, $p < 0.001$, Cohen’s $d=-0.74$). Three families showed modestly higher pseudo-perplexity for SA generation (SH3, WW, PDZ; $d=0.60$ –1.03), though all SA-generated sequences remained well within the range of natural proteins. Across all eight families, the mean pseudo-perplexity was 5.71 for SA-generated sequences versus 5.58 for stored sequences, confirming that an independent model trained on billions of natural sequences considers SA-generated sequences comparably protein-like to natural family

members. Beyond this language-model scoring, we cross-referenced SA-generated substitutions against published deep mutational scanning (DMS) datasets to ask whether the Boltzmann sampling procedure preferentially generates experimentally functional mutations (SI Appendix, Tables S5 and S6). For the PDZ domain (DLG4/PSD-95 PDZ3 (22)), 91.7% of 4,313 matched substitutions were classified as experimentally tolerated, compared to an overall tolerance rate of 77.5% in the DMS dataset ($1.18\times$ enrichment, binomial $p < 10^{-134}$). For the SH3 domain (GRB2 SH3 (23)), the enrichment was even stronger: 90.4% of 2,396 matched substitutions were tolerated versus 67.5% overall ($1.34\times$ enrichment, binomial $p < 10^{-155}$). In both families, the continuous DMS fitness scores of SA-generated substitutions were significantly higher than those of random single mutations (Wilcoxon $p < 10^{-20}$; SI Appendix, Tables S5 and S6). These results demonstrate that the Boltzmann sampling procedure preferentially generates substitutions at positions and to residues that are experimentally functional, without any explicit fitness objective.

Finally, the beta defensin family (PF00711) provided an opportunity to test whether SA generation preserves not only sequence-level statistics but also the biophysical properties that underpin antimicrobial function (SI Appendix, Fig. S8). SA-generated sequences preserved the six-cysteine disulfide scaffold (6.01 ± 0.08 mean cysteines, compared to 6.02 ± 0.15 for natural sequences), maintained a cationic charge profile ($+6.1 \pm 1.6$) with the narrowest charge distribution of any method, and occupied the same region of charge-hydrophobicity space as natural defensins. By a minimal antimicrobial plausibility filter (net charge $\geq +2$, hydrophobic fraction in $[0.3, 0.7]$, ≥ 4 cysteines), 51% of SA-generated sequences passed, compared to 42% of stored sequences, 25% of HMM-emitted sequences, and 16% of MSA Transformer sequences. This implicit enforcement of biophysical constraints arises because the energy landscape concentrates probability near the stored patterns, which already satisfy these constraints by virtue of being functional proteins; no explicit objective for charge, hydrophobicity, or disulfide topology is required.

Discussion

Across eight Pfam families spanning a ten-fold range of sizes ($K=37\text{--}420$) and a seven-fold range of sequence lengths ($L=23\text{--}262$), SA generated novel sequences that preserved family-level amino acid statistics and were predicted by independent structure predictors to adopt the correct fold, using only the seed alignment and no training, external data, or GPU resources. No learned baseline achieved this combination. Profile HMMs and the MSA Transformer generated sequences with identity well below the within-family range (30–70%), indicating that their high novelty reflected compositional drift rather than meaningful exploration. EvoDiff required millions of pretraining sequences and hours of compute, and collapsed on the longest family. No baseline simultaneously stayed within the family identity range, maintained low composition divergence, and produced genuine novelty. That a training-free method matched systems built on orders of magnitude more data indicated that the statistical structure of protein families can support generation without learned representations, provided the generative framework respects the geometry of the family's sequence space. Beyond matching their output, SA avoids the pretraining corpora, GPU resources, and large parameter counts that place deep generative methods out of reach for the small, single-laboratory families that make up most of Pfam, requiring only a seed alignment and a standard workstation. The consensus-with-noise and column-permuted alignment controls confirmed that this was not a consequence of marginal statistics: SA produced a different output profile from isotropic perturbation of the consensus, and exploited inter-position correlations that column-independent models could not capture.

SA-generated sequences achieved higher TM-scores to the canonical family fold in six of eight families by ESMFold, corroborated by AlphaFold2 across all eight families with TM-score concordance reaching $r=0.999$ (SI Appendix, Fig. S4). pLDDT scores were indistinguishable from stored sequences in six of eight families, confirming that the sampler did not sacrifice folding confidence for diversity, and the single exception (SH3) was resolved by AlphaFold2 cross-validation. Two explanations for the elevated TM-scores merit consideration. First, predictor bias: structure predictors trained on evolutionary data may assign higher confidence to consensus-proximal sequences. Second, genuine structural convergence: the Boltzmann ensemble averages over lineage-specific deviations, potentially producing sequences that encode the shared

fold more cleanly than individual natural sequences. These explanations are not mutually exclusive, and distinguishing them requires experimental structure determination. The DMS validation provided partial evidence for the latter: if elevated TM-scores reflected only predictor bias, one would not expect SA substitutions to be enriched for experimentally tolerated mutations, yet the 1.18–1.34× enrichment in two independent DMS datasets suggested that the sequences encode functional proteins. However, all computational validators used here (structure predictors, ESM2-650M, and DMS-derived fitness scores) were trained on evolutionary data, so high scores for family-like sequences are expected; experimental characterization would provide definitive confirmation.

The WW scaling study showed that generation quality was stable down to $K=20$ sequences, with KL divergence below 0.02 and novelty above 0.47, and replication across three additional families confirmed generality. This robustness arose because the energy landscape is defined entirely by the stored patterns, with no parameters that could overfit to a small training set. The entropy inflection criterion adapted the operating temperature to the memory matrix at each scale, and β^* could be predicted from PCA dimensionality alone ($R^2=0.97$), enabling fully automatic operation from a seed alignment. The $\sqrt{d}$ scaling of β^* followed from concentration of measure on the unit sphere: cross-similarities between stored patterns have variance $1/d$, so the inverse temperature required to resolve individual patterns must grow as $\sqrt{d}$. The reduced coefficient (0.28 vs. ~ 1.6 for random patterns) reflected self-similarity anchoring: each stored pattern has unit inner product with itself but only $\mathcal{O}(1/\sqrt{d})$ similarity to other patterns, so the softmax concentrated on the self-pattern at lower β than a mean-field model predicts (SI Appendix, Section S19). By contrast, learned generative models face a parameter-to-data ratio problem when $K < 100$: VAEs and diffusion models have thousands to millions of parameters that cannot be constrained by a few dozen sequences. This regime encompasses most Pfam families.

The connection between attention and Boltzmann sampling extends beyond proteins: any set of examples representable as vectors defines an energy landscape via the modern Hopfield energy, and Langevin dynamics on this landscape provides a generative model with no training. Two conditions determine whether the approach produces useful output. First, the stored examples must be structured enough that the energy landscape has well-defined basins rather than a featureless plateau; protein families satisfy this because evolutionary constraint creates correlated patterns of conservation and variation. Second, the examples must be diverse enough that the sampler does not trivially retrieve: if all stored patterns are nearly identical, the distribution collapses and generation reduces to consensus with noise. The eight families studied here spanned 22–37% mean pairwise identity, confirming that the method operated in the intermediate regime between these extremes. Classical Hopfield networks were limited to storing $\mathcal{O}(d)$ patterns before catastrophic interference (24, 25). The modern continuous Hopfield energy turns this into an advantage for scarce-data generation: with $K \ll d$ stored patterns, the energy landscape is well-separated and the sampler can explore between memories without interference. The same approach may apply to other biological sequence families where data is scarce but examples are structured, such as antibody repertoires where clonotypes have only a handful of somatic variants, RNA families where Rfam seed alignments are smaller than Pfam counterparts, or small-molecule libraries with tens of analogs per series. The Hopfield energy also admits a conditioning mechanism: adjusting pattern multiplicities via a scalar bias on the attention logits shifts generation toward a user-specified functional subset, enabling property-directed generation (e.g., toward binding activity) without retraining (26). The analytic score function interfaces with the convergence theory of Langevin dynamics (27, 28), enabling formal guarantees on sample quality that are absent from most neural generative models.

Several limitations should be acknowledged. The most significant is that SA operates on a fixed alignment representation: it requires a pre-computed MSA and a fixed alignment length L , and cannot generate insertions, deletions, or variable-length sequences. This constraint is inherited from the one-hot-plus-PCA encoding, which maps each sequence to a vector of fixed dimension $20L$; any change in L would alter the representation space itself. In practice, this means the method is best suited to compact, well-aligned domains rather than multi-domain proteins or families with frequent large insertions. Extending stochastic

attention to variable-length generation would likely require replacing PCA with a length-agnostic embedding while preserving the closed-form score function, a nontrivial but tractable direction for future work. The PCA projection itself discards information that may be relevant for capturing higher-order correlations, though a sensitivity analysis confirmed robustness across the 75%–99% retained variance range (SI Appendix, Table S14). The eight families studied here are all well-structured, globular domains with abundant structural and functional annotation; this choice was necessary for validation but introduced a selection bias toward precisely the class of proteins that structure predictors handle best. Performance on intrinsically disordered regions, transmembrane domains, multi-domain proteins, or families with very low sequence identity remains untested, and the method’s reliance on PCA may be particularly limiting for disordered families where sequence variation is less structured. Finally, our multi-chain protocol does not guarantee coverage of all modes in the Boltzmann distribution, particularly for very large or highly diverse families, and wet-lab characterization of selected generated sequences would be needed to establish the method’s utility for protein engineering applications.

Materials and Methods

Stochastic Attention. Let $\mathbf{X} = [\mathbf{m}_1, \dots, \mathbf{m}_K] \in \mathbb{R}^{d \times K}$ be the memory matrix whose columns are vectorized protein sequences from a family alignment of K members, each represented in d dimensions after one-hot encoding and PCA projection. Each sequence of length L is first one-hot encoded to a binary vector in $\{0, 1\}^{20L}$, then projected by PCA (retaining 95% of variance, yielding $d \ll 20L$) and unit-normalized to form the corresponding column $\mathbf{m}_k$; full encoding, projection, and decoding details are given in SI Appendix, Section S18. The modern Hopfield energy (18) of a state $\boldsymbol{\xi} \in \mathbb{R}^d$ is:

$$E(\boldsymbol{\xi}) = \frac{1}{2} \|\boldsymbol{\xi}\|^2 - \frac{1}{\beta} \log \sum_{k=1}^K \exp(\beta \mathbf{m}_k^\top \boldsymbol{\xi}), \quad [1]$$

where $\beta > 0$ is an inverse temperature. The Boltzmann distribution $p_\beta(\boldsymbol{\xi}) \propto \exp(-\beta E(\boldsymbol{\xi}))$ has score function $\nabla \log p_\beta(\boldsymbol{\xi}) = \beta(\mathbf{T}(\boldsymbol{\xi}) - \boldsymbol{\xi})$, where $\mathbf{T}(\boldsymbol{\xi}) := \mathbf{X} \text{softmax}(\beta \mathbf{X}^\top \boldsymbol{\xi})$ is the softmax attention retrieval map. This score is *exact* and computed by a single attention operation; no score network or training is required. Applying the unadjusted Langevin algorithm (ULA) (29) to the potential $U = \beta E$ with step size α/β , where $\alpha \in (0, 1)$ is the mixing coefficient, yields the SA update:

$$\boldsymbol{\xi}_{t+1} = (1 - \alpha) \boldsymbol{\xi}_t + \alpha \mathbf{X} \text{softmax}(\beta \mathbf{X}^\top \boldsymbol{\xi}_t) + \sqrt{\frac{2\alpha}{\beta}} \boldsymbol{\epsilon}_t, \quad \boldsymbol{\epsilon}_t \sim \mathcal{N}(\mathbf{0}, \mathbf{I}). \quad [2]$$

Each step performs contraction toward the origin, a softmax-weighted pull toward stored memories, and isotropic Gaussian noise, at a per-step cost of $\mathcal{O}(dK)$.

Temperature Selection and Sequence Decoding. The critical inverse temperature β^* is identified via the entropy inflection criterion: we evaluate the Boltzmann entropy $H(\beta) = -\sum_k w_k(\beta) \log w_k(\beta)$, where $w_k(\beta) = \text{softmax}(\beta \mathbf{X}^\top \boldsymbol{\xi})_k$ are the attention weights at a representative stored pattern $\boldsymbol{\xi} = \mathbf{m}_1$, and locate the inflection point $\beta^* = \arg \max_\beta |d^2 H/d\beta^2|$ via a grid search over $\beta \in [0.1, 20]$. Below β^* the sampler visits all patterns equiprobably; above β^* pattern-specific basins emerge. Generation uses $\beta_{\text{gen}} \approx 2\beta^*$, just above the inflection, to balance exploration and fidelity.

To decode a PCA-space sample $\boldsymbol{\xi}$ to an amino acid sequence, the PCA projection is inverted to recover a one-hot-like vector in the original $20L$ -dimensional space, and the amino acid at each position is assigned by argmax over the 20 channels. The resulting length- L sequence is the decoded output; positions where the PCA reconstruction assigns negligible weight to all amino acids default to the most frequent residue in the stored alignment at that position. The complete pipeline is summarized in Algorithm 1.

Experimental Design. We selected eight Pfam families spanning a range of sizes, sequence lengths, and conservation levels (Table 1): RRM (PF00076, $K=68$, $L=71$), SH3 (PF00018, $K=55$, $L=48$), WW (PF00397, $K=420$, $L=31$), Kunitz (PF00014, $K=99$, $L=53$), zf-C2H2 (PF00096, $K=151$, $L=23$), PDZ (PF00595, $K=44$, $L=83$), Pkinase (PF00069, $K=37$, $L=262$), and Defensin_beta (PF00711, $K=45$, $L=36$). Seed alignments were downloaded from InterPro (30) and cleaned by removing columns with $> 50\%$ gaps and sequences with $> 30\%$ gaps. PCA retained 95% of variance, followed by unit-norm projection to form the memory matrix. For each family, we ran 30 independent

Algorithm 1 Stochastic Attention Protein Sequence Generation

Require: Seed alignment $\mathbf{S} \in \mathcal{A}^{K \times L}$, variance threshold $\rho_{\min} = 0.95$, mixing coefficient $\alpha = 0.01$, chains $N_c = 30$, iterations $T = 5000$, burn-in $T_b = 2000$, thinning $\Delta t = 100$, samples/chain $s = 5$

Ensure: Generated sequences $\{\mathbf{g}_1, \dots, \mathbf{g}_S\} \subset \mathcal{A}^L$, $S = N_c \cdot s$

```
1: Encoding:
2: One-hot encode:  $\mathbf{B} \leftarrow [\text{onehot}(\mathbf{S}_{1,:}), \dots, \text{onehot}(\mathbf{S}_{K,:})] \in \{0, 1\}^{20L \times K}$ 
3: Center:  $\bar{\mathbf{x}} \leftarrow \frac{1}{K} \sum_k \mathbf{B}_{:,k}$ ;  $\tilde{\mathbf{B}} \leftarrow \mathbf{B} - \bar{\mathbf{x}} \mathbf{1}_K^\top$ 
4: SVD:  $\tilde{\mathbf{B}} = \mathbf{U} \Sigma \mathbf{V}^\top$ 
5: Select  $d$ : smallest  $p$  such that  $\sum_{j=1}^p \sigma_j^2 / \sum_j \sigma_j^2 \geq \rho_{\min}$ ; set  $\mathbf{W}_d \leftarrow \mathbf{U}_{:,1:d}$ 
6: Project and normalize:  $\mathbf{m}_k \leftarrow \mathbf{W}_d^\top (\mathbf{B}_{:,k} - \bar{\mathbf{x}}) / \|\mathbf{W}_d^\top (\mathbf{B}_{:,k} - \bar{\mathbf{x}})\|$  for  $k = 1, \dots, K$ 
7: Assemble memory:  $\mathbf{X} \leftarrow [\mathbf{m}_1, \dots, \mathbf{m}_K] \in \mathbb{R}^{d \times K}$ 
8: Temperature selection:
9: Find  $\beta^*$ : entropy inflection of  $H(\beta) = -\sum_k p_k \log p_k$  with  $p_k = \text{softmax}(\beta \mathbf{X}^\top \mathbf{m}_k)_k$ 
10: Set  $\beta_{\text{gen}} \leftarrow \max(\lceil 2\beta^* \rceil, 5)$ 
11: Multi-chain Langevin sampling:
12:  $\mathcal{G} \leftarrow \emptyset$ 
13: for  $c = 1, \dots, N_c$  do
14:   Draw  $k_c \sim \text{Uniform}\{1, \dots, K\}$ 
15:    $\xi_0 \leftarrow \mathbf{m}_{k_c} + 0.01 \boldsymbol{\eta}$ ,  $\boldsymbol{\eta} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_d)$ 
16:   for  $t = 0, \dots, T - 1$  do
17:      $\mathbf{a}_t \leftarrow \text{softmax}(\beta_{\text{gen}} \mathbf{X}^\top \xi_t)$ 
18:      $\boldsymbol{\epsilon}_t \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_d)$ 
19:      $\xi_{t+1} \leftarrow (1 - \alpha) \xi_t + \alpha \mathbf{X} \mathbf{a}_t + \sqrt{2\alpha/\beta_{\text{gen}}} \boldsymbol{\epsilon}_t$ 
20:   Collect  $s$  evenly spaced samples from  $\{\xi_{T_b+1}, \xi_{T_b+\Delta t+1}, \dots, \xi_T\}$  into  $\mathcal{G}$ 
21: Decoding:
22: for each  $\xi \in \mathcal{G}$  do
23:   Inverse PCA:  $\hat{\mathbf{x}} \leftarrow \bar{\mathbf{x}} + \mathbf{W}_d \xi \in \mathbb{R}^{20L}$ 
24:   Argmax decode:  $g_\ell \leftarrow \arg \max_{a \in \mathcal{A}} \hat{x}_{20(\ell-1)+a}$  for  $\ell = 1, \dots, L$ 
25: return  $\{\mathbf{g}_1, \dots, \mathbf{g}_S\}$ 
```

347 ULA chains of $T=5,000$ iterations ($\alpha=0.01$), discarding 2,000 burn-in iterations and thinning every 100 steps to retain
348 5 samples per chain ($S=150$ sequences total). The inverse temperature was set automatically via the entropy inflection
349 criterion, with a generation regime at $\beta_{\text{gen}} \approx 2\beta^*$ and a retrieval regime at $\beta_{\text{ret}} \approx 20\beta^*$. The thinning interval of
350 100 steps is well-matched to the integrated autocorrelation time of the Hopfield energy ($\tau_{\text{int}} = 76\text{--}122$ steps across
351 families), yielding approximately 35 effective independent samples per chain and $\approx 1,050$ effective draws total; the
352 burn-in of 2,000 steps provides a $3.2\text{--}6.6\times$ margin over the empirical chain convergence point (SI Appendix, Table S8).
353 Initialization sensitivity was verified by repeating the full pipeline with random sphere initialization ($\xi_0 = \mathbf{z}/\|\mathbf{z}\|$,
354 $\mathbf{z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_d)$) in place of stored-pattern initialization; output metrics were indistinguishable across all eight families
355 (maximum absolute KL difference 0.0003), confirming that the generated ensemble reflects the Boltzmann distribution
356 rather than an artifact of starting conditions (SI Appendix, Table S9).

357 To contextualize SA against methods that require pretraining on large external corpora, we compared four
358 established baselines: (i) profile HMM emit via HMMER3 (5); (ii) EvoDiff MSA-conditioned diffusion (15), pretrained
359 on UniRef50 MSAs; (iii) MSA Transformer (14) via iterative Gibbs-like masked language model sampling; and (iv) a
360 Potts model fitted via pseudo-likelihood maximization (plmDCA) (8). For each baseline, the same 150-sequence budget
361 and evaluation metrics (KL divergence, novelty, sequence identity) were applied. Structural plausibility was assessed
362 by predicting folded structures for 50 sequences per method per family using ESMFold (21) and AlphaFold2 (1) (via
363 ColabFold (31)), comparing each predicted structure to an experimentally determined reference using TM-align (32).
364 All statistical comparisons used Wilcoxon rank-sum tests with Benjamini-Hochberg false discovery rate (FDR)

correction at $q=0.05$.

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Data, Materials, and Software Availability

All code, data, seed alignments, generated sequences, and analysis scripts are publicly available at <https://github.com/varnerlab/SA-Protein-Modeling-Study> (20). Pfam seed alignments were obtained from InterPro (30). No new experimental data were generated for this study; all validation used publicly available computational tools (ESMFold, AlphaFold2 via ColabFold, ESM2-650M) and published deep mutational scanning datasets.

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Table 1. Properties of the eight Pfam families used in this study. K : number of seed sequences after cleaning. L : alignment length. d : PCA dimension (95% variance). $\bar{H}_{\text{col}}$: mean per-column Shannon entropy (nats). K_{eff} : effective number of sequences at 80% identity. β^* : empirical entropy inflection point.

Family	Pfam ID	K	L	d	$\bar{H}_{\text{col}}$	K_{eff}	β^*	$\beta^* / \sqrt{d}$
RRM	PF00076	68	71	59	1.92	68	3.85	0.50
SH3	PF00018	55	48	46	1.68	55	3.85	0.57
WW	PF00397	420	31	186	1.74	419	5.45	0.40
Kunitz	PF00014	99	53	80	1.61	99	3.85	0.43
zf-C2H2	PF00096	151	23	106	1.91	151	4.58	0.44
PDZ	PF00595	44	83	37	1.88	43	3.23	0.53
Pkinase	PF00069	37	262	34	1.72	37	3.23	0.55
Defensin_beta	PF00711	45	36	37	1.65	45	3.23	0.53

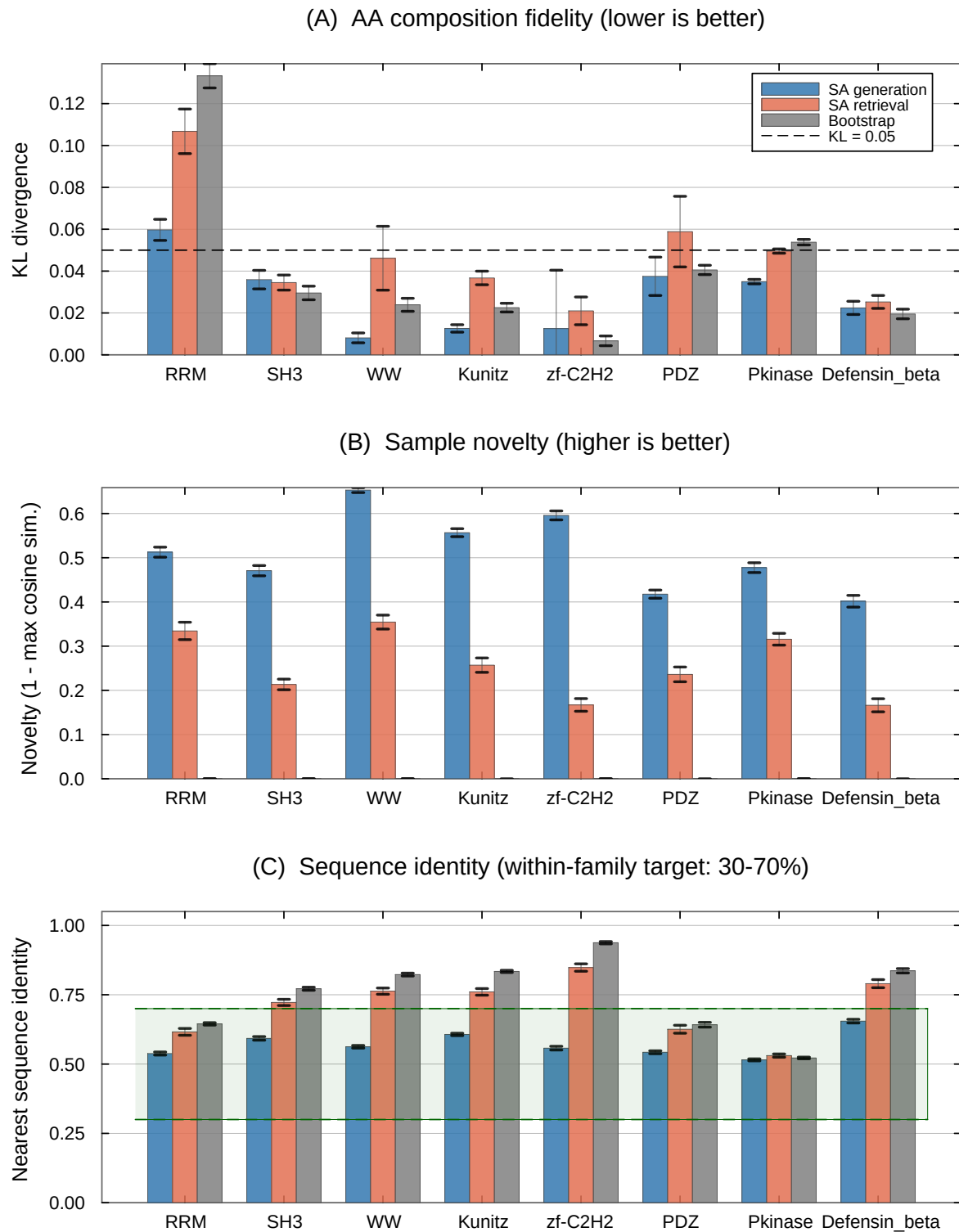


Fig. 1. Cross-family comparison of SA generation, SA retrieval, and bootstrap across eight Pfam families. **(A)** KL divergence of amino acid composition (lower is better); dashed line at $KL=0.05$. **(B)** Novelty in PCA space ($1 - \max_k \cos(\hat{\xi}, \mathbf{m}_k)$; higher is better). Bootstrap samples have zero novelty by construction. **(C)** Nearest sequence identity to a stored pattern; green shaded band (30–70%) marks the typical within-family identity range. Error bars: ± 1 SE (30 chains $\times$ 5 samples).

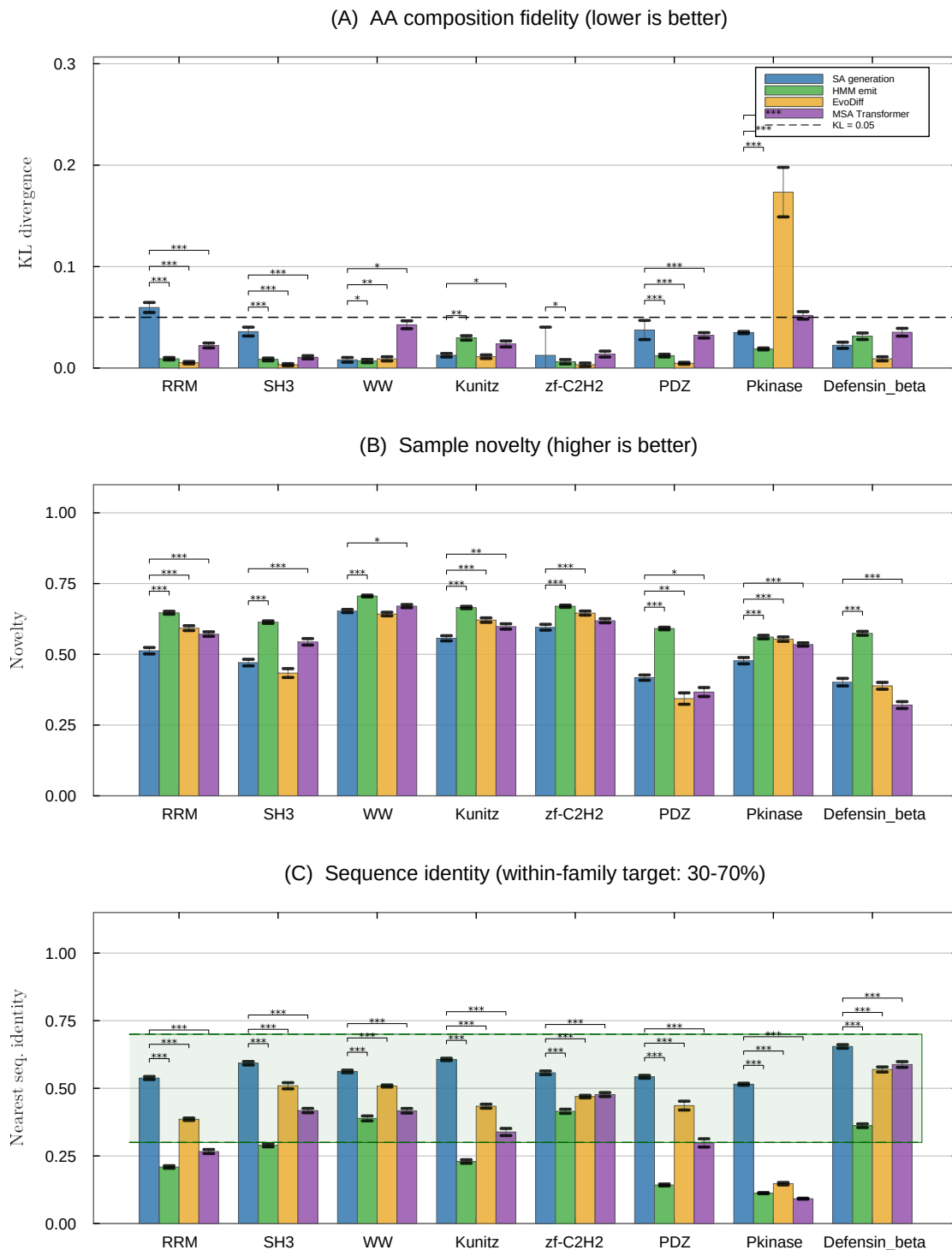


Fig. 3. Head-to-head comparison of SA generation against three learned baselines across all eight Pfam families. **(A)** KL divergence of amino acid composition (bootstrap SE; $n_{boot}=1000$); dashed line at $KL=0.05$. **(B)** Novelty in PCA space. **(C)** Nearest sequence identity to a stored pattern; green shaded band (30–70%) marks the within-family identity range. Panels (B) and (C): per-chain means ± 1 SE. Significance brackets: Wilcoxon rank-sum tests ($***p < 0.001$, $**p < 0.01$, $*p < 0.05$). Per-family numerical values are reported in SI Appendix for the learned baselines (Table S2) and for SA generation and the reference conditions (Table S1).

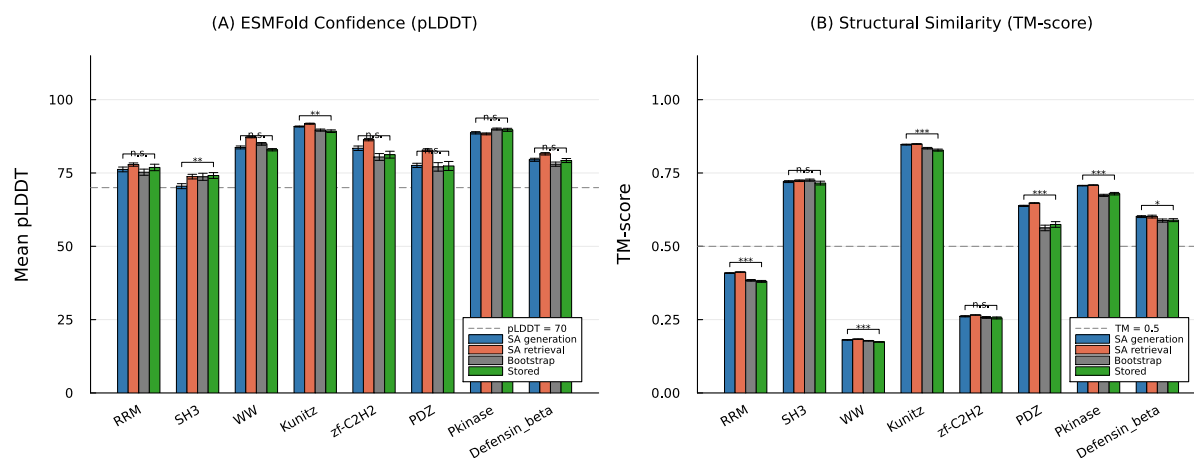


Fig. 4. Structure validation using ESMFold. **(A)** Mean pLDDT across all eight families for SA generation, SA retrieval, bootstrap, and stored sequences; dashed line at pLDDT=70. **(B)** TM-score to the experimentally determined reference structure; dashed line at TM=0.5. Significance: Wilcoxon rank-sum tests (*** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$, n.s. = not significant). Error bars: ± 1 SE ($n=50$).

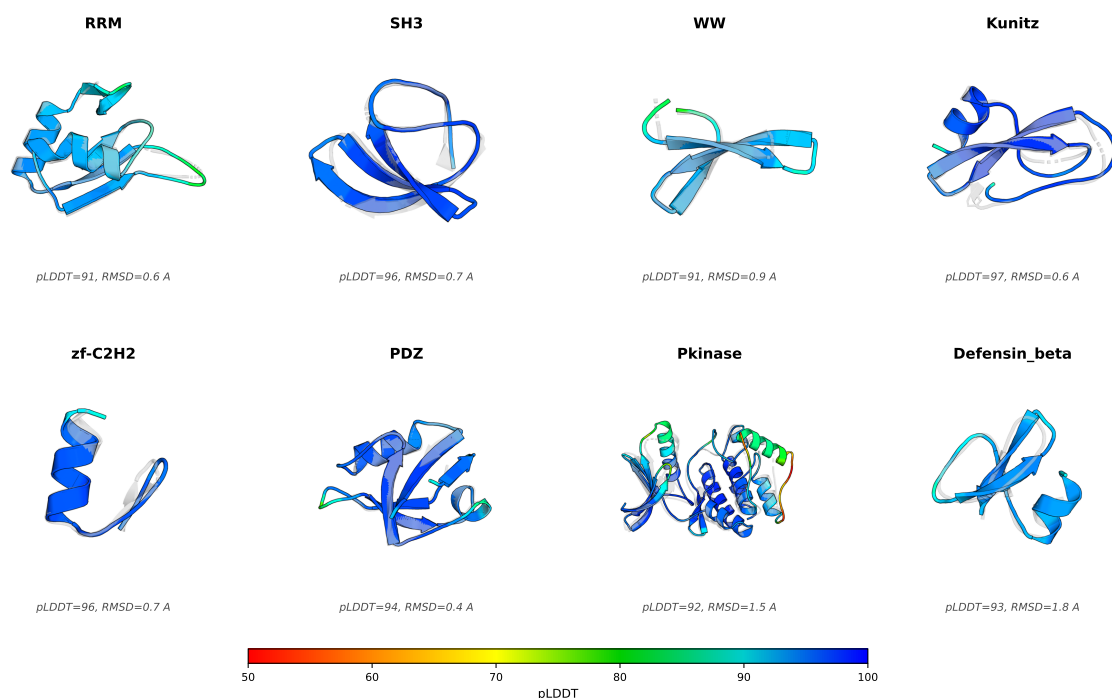


Fig. 5. AlphaFold2 predicted structures for representative SA-generated sequences (colored by per-residue pLDDT confidence) superimposed on the experimentally determined reference structure for each family (gray, semi-transparent). For each family, the SA-generated sequence with the highest mean pLDDT was selected and aligned to the reference using PyMOL. The close predicted structural agreement (sub-angstrom root-mean-square deviation (RMSD)) and uniformly high pLDDT scores are consistent with SA-generated sequences encoding three-dimensional folds that recapitulate the canonical architecture of their target family, despite 40–65% sequence novelty.